

Bulk Proxy Forge — Process & Design

COM/CSC 271

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Part II: Identifying the Product & Target Audience

A. Define Your Product

1. What problem does your product website solve or address?

Creating a large number of custom Magic: The Gathering proxies is unnecessarily fragmented and repetitive. A user may already have a complete deck list in Scryfall and custom artwork ready to use, but still has to move between several tools to rebuild card data, apply frames, fit rules text into the correct areas, assign artwork and set symbols, create different card backs for different decks, and prepare hundreds of correctly paired front and back image files for a printing service.

In my own recent project, I prepared roughly 500 cards, including custom art for more than 300 of them. Building and checking the card frames and finished images took about 10–20 hours. Preparing the final print order manually would have required another 2–5 hours of renaming, duplicating, matching, and organizing files. The individual tasks are not especially difficult, but repeating them hundreds of times creates a large time cost and many opportunities for mistakes. Also worth noting that I did not do each card manually. I created rules and wrote a program to apply those rules to all of the cards. That took 10–20 hours. If I had done it all manually, it would have taken substantially longer.

2. What is your product idea? How does your website support this idea and help solve the problem?

Bulk Proxy Forge is a web-based application that starts with existing MTG deck lists and ends with one bulk order ready to be printed by a printing service. A user begins by pasting a public Scryfall deck link. Instead of importing only the card names, Bulk Proxy Forge uses the selected printing and associated card data, including artwork, artist, rules text, set, collector number, card layout, and other metadata.

The user can then choose the artwork source, card templates, set symbols, and card back for each deck. Cards are generated from reusable templates instead of requiring the user to rebuild frames and position every text box manually. Multiple decks can remain separate for customization while still being combined into one final print order. The application then pairs each card front with the correct card back and builds the folder structure needed by the printer.

For someone who already has the deck lists and artwork, the goal is to reduce a process that can take 15–25 hours of partially automated work or much more if done completely manually to roughly five minutes of user decisions plus automated processing time.

3. What type of website are you building?

Bulk Proxy Forge is primarily a web application / service website. It combines the explanatory role of a product website with an interactive tool that lets users import decks, configure proxy options, review generated cards, and prepare a print order.

4. Mission statement

Bulk Proxy Forge helps Magic: The Gathering players turn existing deck lists and custom artwork into print-ready proxy orders quickly and accurately by automating repetitive card setup, deck-specific customization, front/back pairing, and print-order preparation.

B. Profile of Target Audience

1. Primary audience

Demographics

- Age range: approximately 10–45, with the core audience expected to be adults who actively play tabletop Magic and are comfortable managing deck lists online.
- Gender distribution: all genders. The interface and branding are not designed around a gender-specific audience.
- Country: initially focused on users in the United States because the prototype is being designed around English-language card data and commonly used online printing workflows, but the core concept could support international users.
- Urban/rural: both. Because the service is web-based, location density is not important as long as the user has internet access and access to a printing/shipping service.

Socioeconomic details

- Average income: approximately \$30,000–\$100,000 as a planning estimate, with a wide range from college students to established working professionals.
- Education: any; no specialized education is required.
- Marital/family status: varied and not central to the product.
- Occupation: students, software/technical workers, office professionals, tradespeople, hobbyists, and other working adults.
- Hours worked per week: typically 0–50 depending on whether the visitor is a student, part-time worker, or full-time professional.

Web behavior

- Web use: Irrelevant as long as they have a computer or phone and are willing to use it for 30-60 minutes while creating and ordering the product.
- Devices: desktop/laptop is the primary device being designed for because it will be easier with more screen space for this application. However, it should still be easy to use on mobile.

2. Fictional visitor chart

Name	Sex	Age	Location	Occupation	Income	MTG relationship
Alex Morgan	Male	29	Boston, MA	Software developer	\$92,000	player who is willing to pay for normal cards, but not cards on the restricted list because they cost over \$1000 each.
Maya Chen	Female	24	Providence, RI	Graduate student	\$32,000	New casual player who wants to try playing with friends, but can't afford \$500+ for real cards. A friend helped her build a deck she likes on Scryfall.
Jordan Ellis	Male	38	Columbus, OH	Electrician	\$68,000	Regular CEDH player who plays every weekend at a local shop.

3. Reasons people would visit the site and visitor tasks

Alex Morgan

- Least likely of the 3 to use the site because he can reasonably afford the cards, so social pressure to buy real ones may outweigh the cost savings if he just wants cards to play with.
- He may want custom art or to make a nice looking, themed deck where the whole deck looks like a single coherent product which would make templates attractive to him.
- He may create a custom art deck for a friend/spouse focused more on how they look as a nice, personalized gift.
- Not very likely to order in bulk. The main appeal of the service is the simplicity and time savings on his busy schedule.

Maya Chen

- Her limited budget forces her to either not play at all or buy proxies.
- She is unfamiliar with the game, so likely to flake if the process isn't simple for someone with no prior knowledge/experience.
- She would take the decklist her friend made for her and import it to the site and gloss over any settings, just looking for the "finish it" or "complete order" type of button, not necessarily understanding the difference between the card database site, our site, and the printing site.
- For her, the guided process and loading/status bars would be extremely important so she doesn't get lost.

Jordan Ellis

- He is a hardcore MTG fan who likes to build new decks, but can't afford to spend tons of money on restricted list cards. He is already used to making proxies, and uses another similar service, but he has to choose between having custom card backs for his decks or ordering them separately (costing more money) or doing a huge amount of manual work to create a bulk order with different backs.
- He is our primary target audience. He wants to use the site regularly. He wants to make his own templates. Have many decks being created/imported regularly, and wants to buy big bulk orders to save on the cost of printing.
- He wants to make fully customized decks with custom art he acquired from an AI image generator, and proxies all of his decks. Our site saves him a huge amount of time and effort.

Part III: Planning & Designing the Product Website

A. Website Requirements

1. Purpose & Goals

Primary goal

The primary goal of Bulk Proxy Forge is to reduce the amount of repetitive work required to turn an existing MTG deck and artwork into finished proxy images and a correctly organized print order.

Most important user action

The most important first action is for the user to import a deck, ideally by pasting a Scryfall deck link. That gives the application the card list and exact printings needed to drive the rest of the workflow.

A successful user journey is:

Import deck → (optionally) choose art/style/back → generate cards → review front/back pairs → build print order and end on the review/checkout page of a card printing service.

2. Content & Features

Critical pages and features

- Home / Get Started — ~1 minute tutorial video imbedded that is optional. This page needs to be very sparse. We want just a couple of text boxes, and an “Import Deck” button with some minimal amount of description on how to use it in case they don’t want to watch the video. There is only 1 option for a new user. “Import a deck”. They don’t need to know anything else yet. They can see something like progress steps like “Import Deck” -> “Choose Art/Style” -> “Review and Checkout”. We should have 1 other button that allows them to go to the main menu. That is all. No extra clutter on the new user page.

For return users:

- Deck Library — shows the user’s imported decks and their current preparation status.
- Template Forge — Place for building custom templates.
- Import Deck — accepts a Scryfall deck link or card list and creates a new deck workspace. (located in the Deck Library and home page).
- Deck Viewer — shown when clicking on a deck in the Deck Library. Shows all cards in a deck and the options/controls for that deck.
- Card Editor / Inspector — lets the user view an individual card, and override individual card settings or template.
- Print Orders — lets the user select one or more completed decks, review paired fronts and backs, and create the final order package. (located in 2 places, in the Deck Library when cards are selected, and at the end of the process of importing/generating a new deck that asks “Checkout or return to Main Menu?”).
- Discord button — For help/community.
- Settings — account level settings that persist across all decks.

Homepage highlights

The homepage should immediately communicate three things:

1. What the product does: turns Scryfall decks and artwork into print-ready proxies.
2. Why it matters: saves hours of repetitive setup and file organization.
3. What to do next: paste/import a deck.

The first screen should include the Bulk Proxy Forge logo, a short one-sentence value proposition, an Import Deck button, and a simple three-step visual:

1. Import a deck
2. Choose art and style
3. Review and checkout

3. Look & Feel

Design inspiration

- Extreme simplicity — We want people who know nothing about magic to be able to take a deck link, import, use default settings, and order without feeling stressed or confused.
- Forge — Style of the website will be heavily inspired by a metal forge, so dark metallic + orange.

Overall style

The site should feel modern, dark, focused, and slightly fantasy/industrial without becoming visually noisy. The visual theme is a forge: black and charcoal backgrounds, warm brown metal tones, bright orange highlights, and glowing ember-like accents. The interface should still prioritize readability and obvious navigation over decorative effects.

Color, fonts, and imagery

- Primary background: near-black / charcoal.
- Secondary surfaces: dark brown and forged-metal tones.
- Accent: bright forge orange.
- Main text: warm cream/off-white.
- Success/ready status: muted green.
- Warning/error states: yellow-orange and red.
- Interface font: clean sans-serif for readability.
- Brand/display font: stronger fantasy or serif treatment used sparingly for headings and the logo.
- Imagery: finished proxy cards, card backs, card-frame examples, and how to video.

B. Design & Prototyping

1. SDLC Approach & Timeline

Chosen model: Agile

Agile is the best fit because the product can be built in small usable pieces, tested, and revised throughout the semester. The exact workflow and interface will become clearer as the HTML, CSS, and JavaScript portions of the course are completed, so a rigid Waterfall process would make unnecessary assumptions early in the semester.

Planned timeline

Iteration 1 — Requirements & User Flow Planning

- Identify all required features and rank them by importance.
- Group related features and determine which should be available to new users, returning users, or both.
- Plan the main user flows, including first-time setup and returning-user workflows.
- Determine which functions should eventually be directly visible, grouped onto pages, or placed inside menus.
- Milestone: complete feature list, priorities, functional groups, and user-flow plan.

Iteration 2 — Development & Testing Workspace

- Create a simple internal development page containing basic buttons, forms, and links for running each feature independently.
- Do not focus on styling or final page layout at this stage.
- Use this page as a testing workspace so individual features can be built and tested without depending on the final interface.
- Milestone: development workspace can launch and test each major feature as it is created.

Iteration 3 — Core Feature Development

- Build features one at a time, beginning with the functions most important to the product.
- Examples include Scryfall deck importing, card-data retrieval, artwork selection, template application, set-symbol handling, and card-back assignment.
- Test each feature independently in the development workspace before moving to the next major feature.
- Revise requirements when testing reveals a better approach.
- Milestone: all core card-generation and deck-configuration features function correctly.

Iteration 4 — Print Workflow & Feature Integration

- Connect the completed features into the full workflow.
- Add multi-deck organization, front/back pairing, print-order creation, ZIP generation, review tools, and printer-transfer behavior.
- Test complete workflows for both new and returning users.
- Fix problems with feature interaction before spending significant time on visual design.
- Milestone: the complete Bulk Proxy Forge workflow functions from deck import through print-order preparation.

Iteration 5 — Interface Structure & Wireframe Testing

- Once the required functionality is stable, create simple wireframes showing where features should appear in the final interface.
- Organize features into pages, menus, tabs, and directly visible controls based on the previously tested user flows.
- Test common navigation paths to determine whether the organization feels natural and efficient.

- Reorganize the interface where necessary before detailed styling begins.
- Milestone: final page structure and navigation flow are established.

Iteration 6 — Visual Design, Responsive Layout & Accessibility

- Apply the Bulk Proxy Forge visual identity, including typography, colors, panels, spacing, icons, and forge-themed styling.
- Build responsive layouts for desktop, tablet, and mobile use.
- Add accessibility features such as keyboard navigation, visible focus states, semantic headings, labels, alt text, sufficient contrast, and clear error messages.
- Run WAVE and Google Lighthouse audits and perform manual keyboard and zoom testing.
- Milestone: the application is visually complete, responsive, and accessible.

Iteration 7 — Final Testing & Release

- Test the complete application using realistic decks and workflows.
- Fix remaining bugs, confusing interactions, accessibility problems, and visual inconsistencies.
- Incorporate user feedback without changing stable functionality unnecessarily.
- Prepare the final deployment and presentation.
- Milestone: final website submission.

Visual timeline

Requirements & User Flows → Development Workspace → Core Features → Feature Integration & Print Workflow → Wireframes & Navigation Testing → Visual Design / CSS / Accessibility → Final Testing & Release

Because the project uses an Agile approach, testing and feedback occur during every iteration rather than only at the end. Features are developed in priority order and tested independently before being integrated. Detailed interface design is intentionally delayed until the important functionality and user flows are understood, reducing the chance of spending significant time redesigning polished pages after the underlying workflow changes.

2. Original Logo



3. Site Map

Site map draft

(New users initial page)

Import Deck (accessed from home or Deck Library)

- ├ Deck link
- ├ Art/Style choices
- ├ Building cards
- ├ Review cards
- └ Checkout vs Home page option

(Returning users initial page)

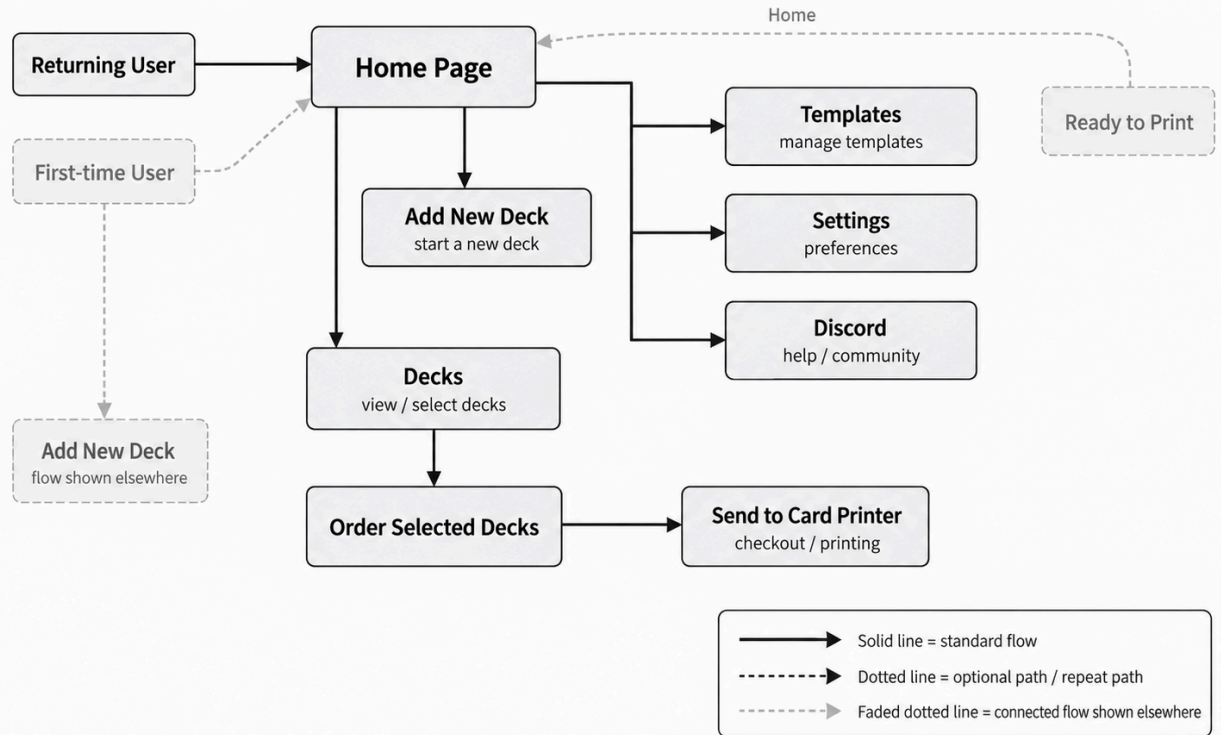
HOME

- ├ Import Deck
- ├ Deck Library
 - ├ Import Deck
 - └ Deck Viewer
 - ├ Cards
 - └ Card Editor
- ├ Templates
- ├ Discord
- └ Settings

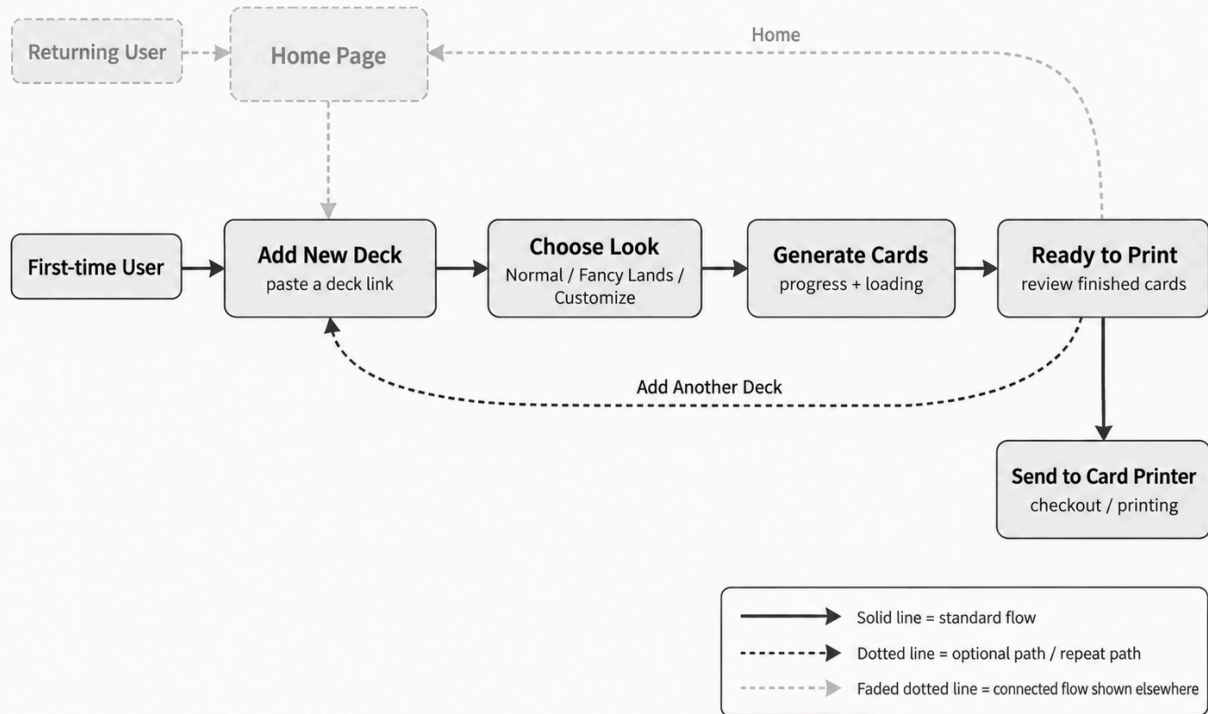
Navigation strategy

The main navigation remains consistent throughout the application. A user can always return to the main page, Deck Library, Templates, or import deck. Within a deck, secondary tabs keep Cards, Art & Setup, and Review grouped together. The normal task flow moves forward from importing a deck to configuring it, reviewing it, and then creating the print order, but the user can move backward at any time to make changes.

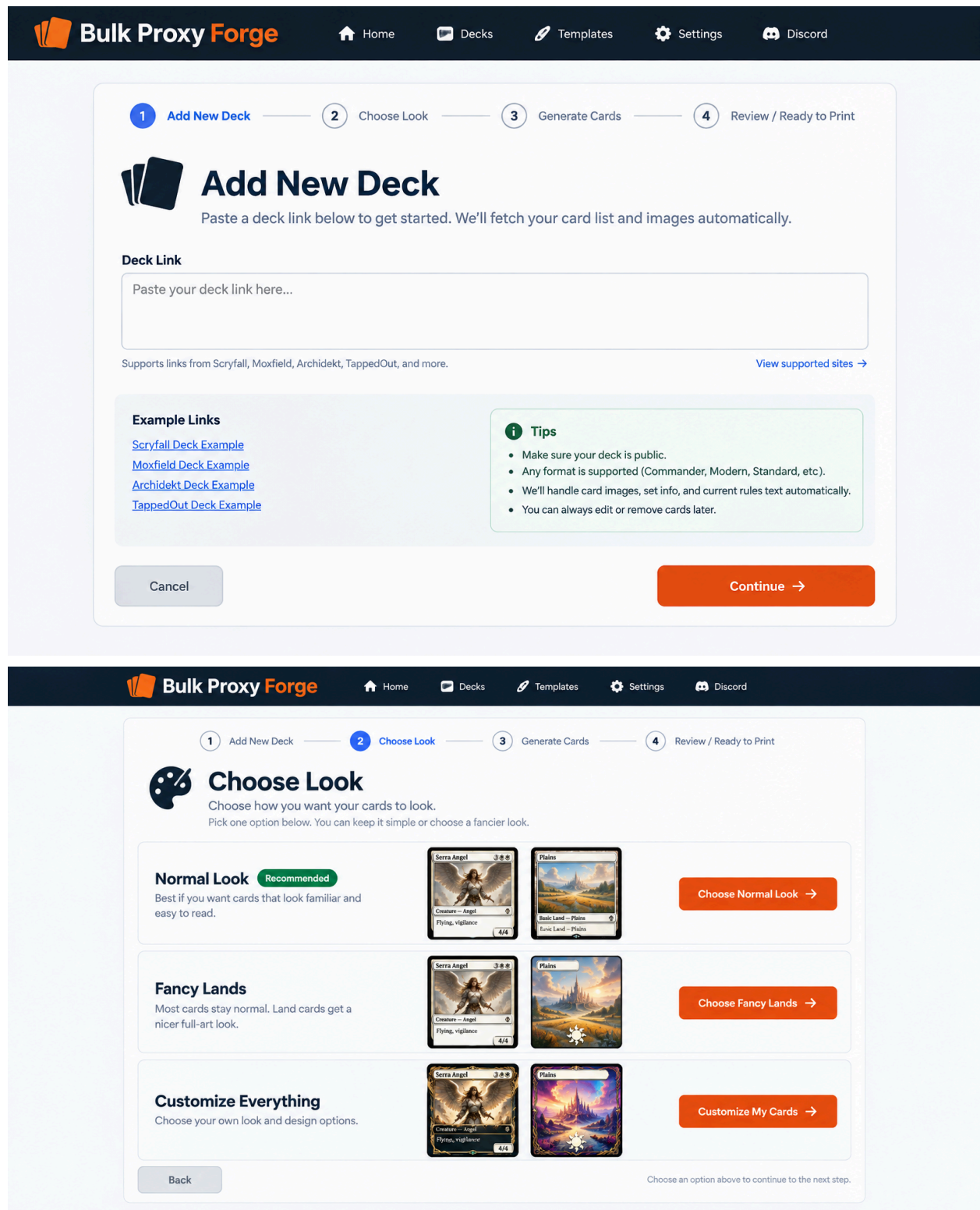
Bulk Proxy Forge — Home Page Flow

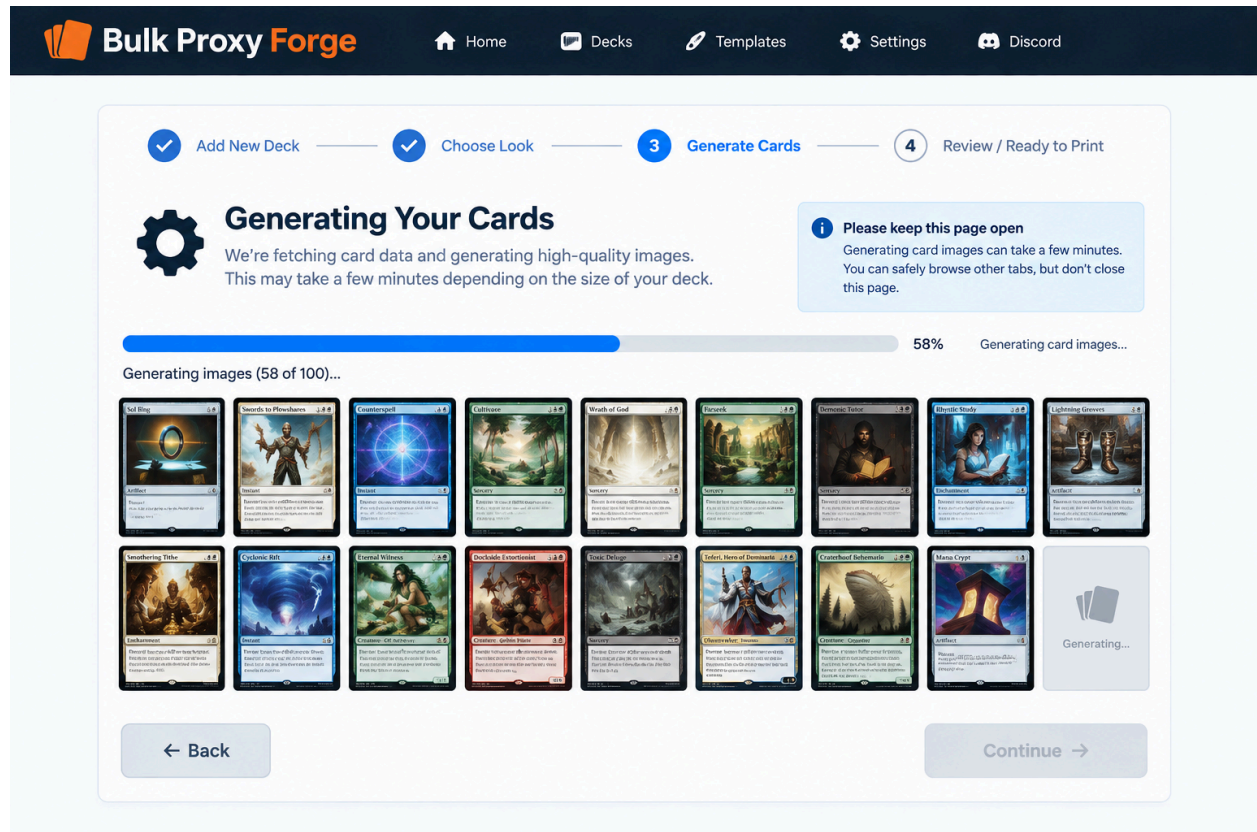


Bulk Proxy Forge — New User Flow



4. Wireframes





5. Accessibility

Bulk Proxy Forge will treat accessibility as part of the initial design rather than something added after the interface is complete.

Visual impairments

- Maintain strong contrast between text and backgrounds.
- Use readable font sizes and allow browser zoom without breaking the layout.
- Use semantic heading levels so screen-reader users can navigate the page structure.
- Provide meaningful alternative text for instructional images and card examples when the image communicates information not already provided in text.
- Avoid using color as the only indicator of status; for example, warnings will use an icon/text label in addition to amber color.

Hearing impairments

- Any video tutorial or demonstration added to the site will include captions or a written transcript.
- Important instructions will always be available as text and will not depend on audio.

Motor impairments

- All major interactions will be usable with a keyboard.
- Interactive controls will have visible focus states.
- Buttons, checkboxes, and other targets will be large enough to select easily.
- The core workflow will not require drag-and-drop as the only way to complete an action.

Cognitive impairments

- Use consistent navigation and terminology throughout the site.
- Break the workflow into clear steps: Import, Art & Style, Generate, Review, Print.
- Keep advanced settings hidden until needed so the main workflow is not visually overwhelming.
- Use clear error messages that explain what happened and what the user should do next.
- Use confirmation screens for destructive or high-impact actions.

Accessibility audit tools and testing

- WAVE for automated accessibility checks.
- Google Lighthouse accessibility audit.
- Manual keyboard-only navigation testing.
- Browser zoom testing at increased magnification.
- Manual review of heading order, form labels, alt text, focus indicators, and contrast.

Submission Asset Checklist

Before submission, the following files will be exported and placed in the documentation folder:

- richards_design.pdf
- logo.png — transparent Bulk Proxy Forge logo
- site_map_new_user.png
- site_map_returning_user.png
- wireframe_new_user_page.png
- wireframe_choose_look.png
- wireframe_generating_cards.png

All files will then be committed and pushed to the course GitHub repository before the repository URL is submitted in Brightspace.